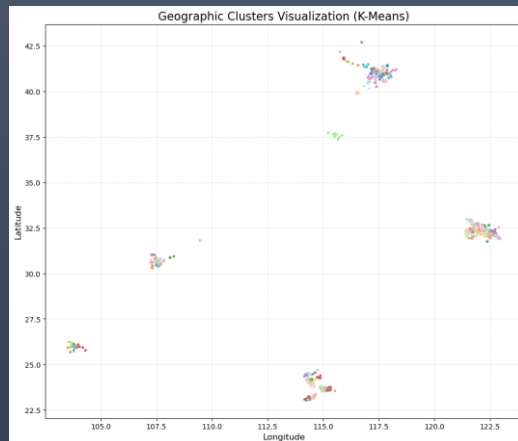


Final Exam

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Data

- 增加经纬度K-means聚类
- 加入文本处理：
 - Jieba分词
 - 去除停用词
 - 人工提取关键词
 - TF-IDF+SVD，分开处理客观描述与主观评价，10+15维
 - 适合树模型，但文本过短影响特征提取
- 编码
 - 数值型、类别型
 - 高基数：Target Encoder
 - 避免维数爆炸、提升运算效率
 - 适合树模型
 - 低基数：独热编码
 - 精确分类



```
# 定义停用词（去除无意义词，减少噪音）
stopwords = {'的', '了', '是', '和', '与', '在', '这', '那', '有', '就', '都', '而'}

def clean_and_cut(text):
    if not isinstance(text, str):
        return ""
    # 去除特殊符号，只保留中文、数字和字母
    text = re.sub(r"[^\u4e00-\u9fa5a-zA-Z0-9]", " ", text)
    # jieba 分词
    words = jieba.cut(text)
    # 去除停用词和短词
    return " ".join([w for w in words if w not in stopwords and len(w) > 1])

keywords_map = {
    'transportation': ['地铁', '城铁', '公交'],
    'is_school_dist': ['学区', '学校', '小学', '中学', '教育'],
    'tax_free': ['满五', '满二', '满两', '唯一', '税费低'],
    'decoration': ['精装', '豪装', '装修'],
    'orientation': ['南北', '采光', '全南'],
    'layout': ['户型方正', '利用率高', '无浪费'],
    'neg_feedback': ['吵', '噪音', '老旧', '偏', '堵', '乱'],
    'pos_feedback': ['安静', '舒适', '素质', '管理好']
}

print("正在提取黄金关键词特征...")
# 将所有文本合并用于关键词搜索
df['all_text_raw'] = df[text_cols].apply(lambda x: ' '.join(x), axis=1)

for feature_name, keywords in keywords_map.items():
    # 如果包含列表中任意一个词，则置为1，否则0
    pattern = '|'.join(keywords)
    df[feature_name] = df['all_text_raw'].str.contains(pattern).astype(int)
```

```
# A. 数值特征
num_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='median'))
])

# B. 高基数特征 -> TargetEncoder (自动平滑)
target_cat_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='constant', fill_value='缺失')),
    ('target_enc', TargetEncoder(target_type='continuous', smooth='auto', random_state=42))
])

# C. 低基数特征 -> OneHot (不产生稀疏矩阵，因为列数很少)
onehot_cat_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='constant', fill_value='缺失')),
    ('onehot', OneHotEncoder(handle_unknown='ignore', sparse_output=False))
])
```

模型优化过程

- XGboost
 - Grid Search, 早停法, 特征工程, 特征筛选, K-fold.....
 - 测试集数据的问题!
- 重新出发
- Base XGboost: 过拟合
- Optuna+CV调参
- K-fold训练模型 (舍弃)
- Permutation Importance筛选特征

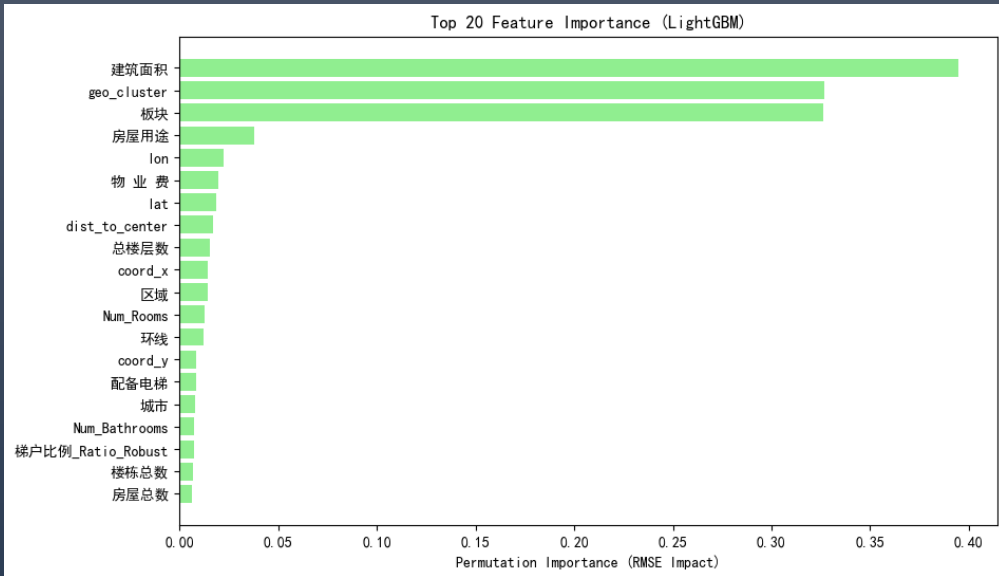
- LightGBM
- 模型Blending

```
[I 2025-12-19 11:53:05,900] Trial 16 finished with  
[I 2025-12-19 11:54:24,935] Trial 17 finished with  
[I 2025-12-19 11:55:30,499] Trial 18 finished with  
[I 2025-12-19 11:56:56,188] Trial 19 finished with  
  
最佳 CV RMSE: 518,717.12  
最佳参数: {'n_estimators': 4703, 'max_depth': 5, 'l
```

```
tree_models = {  
    "XGBoost_GPU": XGBRegressor(  
        n_estimators=5000,  
        learning_rate=0.05,  
        max_depth=6,  
        subsample=0.8,  
        colsample_bytree=0.8,  
        random_state=111, # 你的 RANDOM_STATE  
        n_jobs=-1,  
        device='cuda',    # 使用 GPU  
        tree_method='hist' # 推荐模式  
    )  
}
```

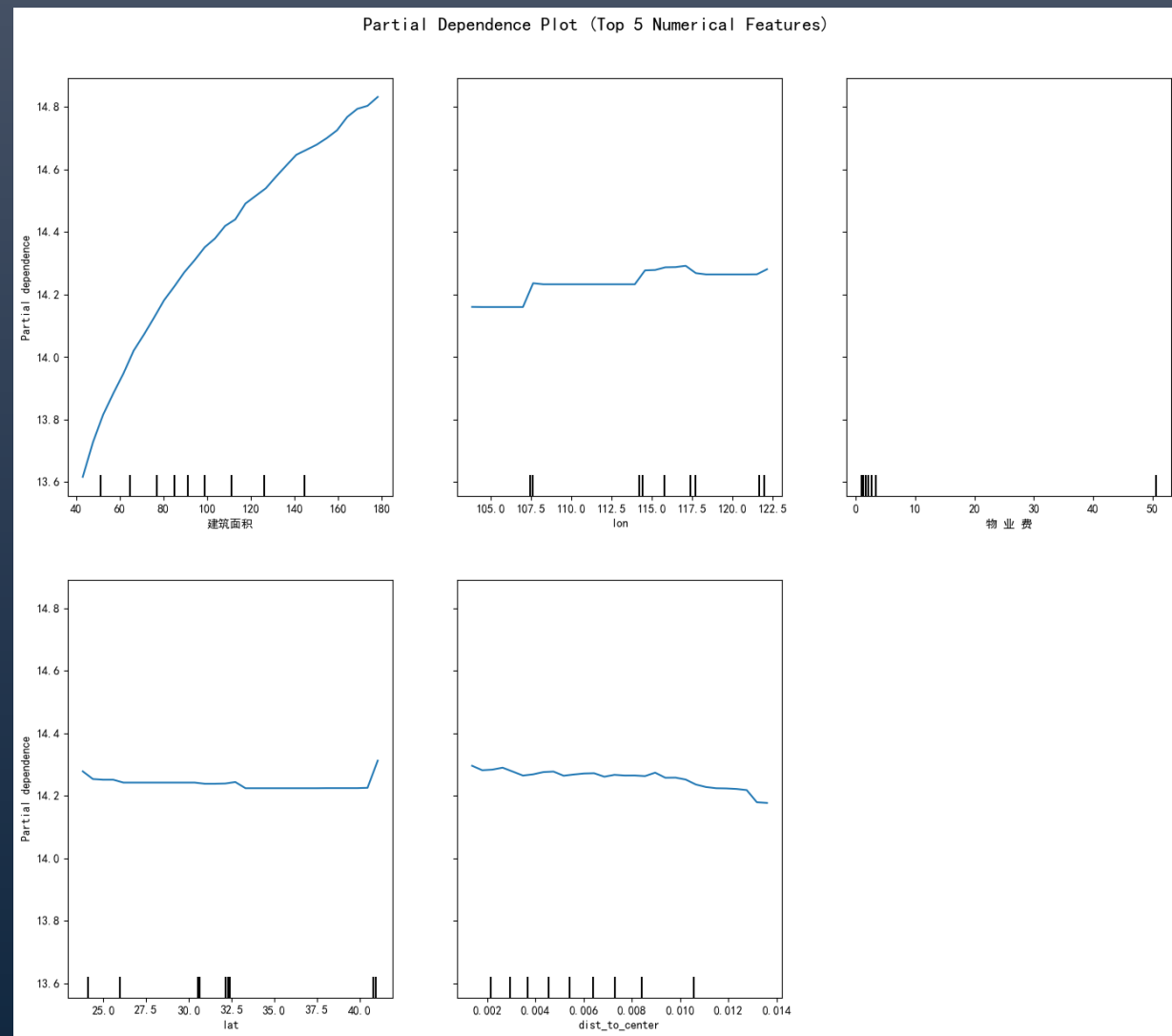
68.3
71.8
64.9
72.6
71.8

	In Sample	Out of Sample	CV	Kaggle
Base	159,343	482,934	521,174	76.8
Optuna	218,579	500,250	518,717	76.7
K-fold		484,208	524,965	76.7
特征筛选	185,099	481,490	521,105	76.8
LightGBM	252,357	476,360	520,455	76.7
Final Model	250,461	475,318	517,829	76.8



🗑️ 建议剔除的特征 (4个):
['ID', 'is_school_dist', 'orientation', 'layout']

	In Sample	Out of Sample	CV	Kaggle
Midterm	853,746	1,004,146	968,678	67.2
	264,650	278,941	276,643	
Base	159,343	482,934	521,174	76.8
	71,997	173,062	168,487	
Final	250,461	475,318	517,829	76.8
	98,971	167,888	161,112	



THANKS!